

Education

Northwestern University B.S. Computer Science & Math	Evanston, IL Expected Graduation: 2029
Lawrence Technological University (Dual Enrollment) 4.0 GPA; Coursework: Multivariable Calculus, Linear Algebra, Intro to Functional Programming, Computer Science 2	Southfield, MI 2026

Experience

Software Development Engineer Intern Tyler Technologies • Built the comprehensive, end-to-end internal police report writing support platform, served SPA in AWS S3 and CloudFront CDN connected to REST APIs with stateless, serverless Lambda endpoints. Formed a highly secure entry gateway using Cognito OIDC, parsing and validating user JWT claims server-side to act as a strict authorization firewall that enforced granular RBAC rules across multi-tenant report infrastructures. • Consolidated API endpoints into feature Lambdas to maximize warm starts, reducing latency during bursts of 1000+ tenant reports from main client; deployed lightweight, flat DynamoDB key-value store to align with stateless nature of AWS Lambda. • Authored modular AWS CDK (TypeScript) infrastructure stacks to automate environment provisioning across Commercial and GovCloud partitions, isolating CJIS-regulated data while maintaining project parity between frontend SPA and backend. • Automated CI/CD workflows using GitHub Actions, enforcing CDK audits, unit tests passing, and security linting on every pull request; including multi-tenant RBAC business logic confirmation in unit tests.	May 2026 – Present
Simons Research Fellow in Computer Science Stony Brook University • Selected as one of 53 fellows out of 1380 applicants (~3.8%) to the Simons Summer Research Program • Leveraged reinforcement learning in parameter space optimization and communication of DAG-based Byzantine fault-tolerant consensus protocols in distributed systems, reducing end-to-end latency of the Autobahn protocol by 49.8% versus default and 73.3% versus random parameter configuration over 6 tested scenarios with different adversarial and normal behavior. • Designed reinforcement learning algorithm with CMAB and Thompson sampling to work within safety/liveness constraints, tested algorithms on Google Cloud Platform setting up virtual machines as nodes across different geographical locations. • Worked with professors from Stony Brook University, University of Pennsylvania, and City University of Hong Kong; submitted pending research at VLDB 2026 (arXiv:2606.09120). Presented a poster and awarded a \$500 stipend.	June 2025 – Mar 2026
Morgan Stanley Finance Academy Morgan Stanley • Selected as one of 150 participants nationwide and instructed by Institutional Securities Group Analysts, applying analytical and problem-solving skills through assignments, case studies, and team-based finance projects.	Oct 2025 – Mar 2026
Blockchain Research Assistant Wayne State University • Programmed and tested smart contracts using Solidity deployed onto the Sepolia Testnet; audited over 10,000 lines of code. • Managed communication between graduate-level classes CSC 8710 and CSC 4996 under Professor Shiyong Lu and assisted in writing and ideating Longevity AI, an AI agent centered around health.	May 2024 – Oct 2025

Selected Projects

The New Page Project – thenewpageproject.com • Solo-built and deployed a React website with animation for a nonprofit, enabling outreach, SEO, and logistics for 20,000+ book donations impacting 680+ students worldwide; maintained reliability and iterated based on user feedback.	2025
AquaNet: Artificial Neural Network for Water Quality Assessment • Developed and trained an artificial neural network to predict water quality indicators in lakes; applied gradient descent optimization and backpropagation algorithms to achieve an average RMSE of 0.3396 over 100 epochs across 3 factors on 100 trials. • Engineered a robust data preprocessing pipeline to clean and normalize large environmental datasets consisting of over 110,000 data points; organized findings into presentable graphs and authored a poster and paper presented at SEFMD and MSEF.	2025

Honors

Regeneron Science Talent Search Scholar (Top 300)	2026
National Merit Scholar (1580 SAT, 800 Math)	2026
American Junior Academies of Science (AJAS) Fellow Gordon & Betty Moore Scholarship Recipient	2025
Wharton Global Investment Competition Semifinalist (Top 50)	2025
NOAA Taking the Pulse of the Planet Award 2nd Place, Science and Engineering Fair of Metro Detroit	2025
Michigan D1 All-State High School Swimming – Team Captain	2023, 2024, 2025, 2026

Technical Skills

Languages: C++, Python, TypeScript (Angular, React), Rust
Cloud/Infrastructure: AWS, GCP, AWS CDK, CloudFormation, S3, CloudFront, Lambda, DynamoDB, Cognito, OIDC, IAM
Tools: Git, GitHub Actions, Docker, Linux (WSL), Agile, CI/CD
AI/ML: PyTorch, pandas, NumPy, Matplotlib